

BMCS3183 Advanced Database Management

Practical Assignment

OVERVIEW

For this assignment you will work in a team of 4 to 5 members to design and develop a database system based on a business scenario using Oracle DBMS with PL/SQL.

Learning Outcomes Being Assessed	• Develop the relational database system with the appropriate integrity constraints and security control. (P3, PLO3) • Design the solutions to issues pertaining to database efficiency and effectiveness using appropriate techniques. (C4, PLO2) • Extract information from the database using efficient SQL query construct. (C4, PLO6)																
Submission Deadline	Monday, 24/08/2026 (Week 11) by 11:59PM. Teams are required to give a script file presentation on Week 8 – 9. Teams are required to give an assignment presentation on Week 11 – 14. For late submission, there will be a reduction of absolute mark's score submitted: • Late 1 to 3 days after deadline of submission: minus 10 marks • Late 4 to 7 days after deadline of submission: minus 20 marks • Late more than 7 days after deadline of submission: 0 marks The report must be type-written using MS-Word (.docx). You are recommended to format your report according to the following specification: <table border="1"> <tr> <td>Media</td><td>Softcopy to be submitted to the <i>course's Google Drive</i>.</td></tr> <tr> <td>Font Size</td><td>PL/SQL code: 9pt Written description : 11pt</td></tr> <tr> <td>Font Style</td><td>Use <i>Times New Roman</i> for body text. Main headings and sub-headings should be clearly stated using suitable font styles (e.g. Arial).</td></tr> <tr> <td>Line Spacing</td><td>Typed material should be <i>1 line spaced</i>.</td></tr> <tr> <td>File & folder Naming Convention</td><td> Digital Submission (Script file, report, PL/SQL codes, etc) to Google Drive: Folder naming format: TeamName (TeamNo - ProgrammeSemGroup) Example: MON0800RMXH (Team 1-RSW2S3G5) Subfolder named as "Member Name" to put an individual works. Report File naming format: TeamName (TeamNo - ProgrammeSemGroup) - StudentFullNamesWithAlphabeticalOrder Example: MON0800RMXH (Team 1-RSW2S3G5) -Alice Ng-Bob Lee-Cat Tan-Dickson Do NOT upload in zip file. </td></tr> <tr> <td>Headers and Footers</td><td>Appropriate footers and headers should be used to enhance clarity and presentation.</td></tr> <tr> <td>Page Numbering</td><td>Ensure that all pages (except cover page) are numbered.</td></tr> <tr> <td>Paper Size</td><td>A4 (210 mm X 297 mm)</td></tr> </table>	Media	Softcopy to be submitted to the <i>course's Google Drive</i> .	Font Size	PL/SQL code: 9pt Written description : 11pt	Font Style	Use <i>Times New Roman</i> for body text. Main headings and sub-headings should be clearly stated using suitable font styles (e.g. Arial).	Line Spacing	Typed material should be <i>1 line spaced</i> .	File & folder Naming Convention	Digital Submission (Script file, report, PL/SQL codes, etc) to Google Drive: Folder naming format: TeamName (TeamNo - ProgrammeSemGroup) Example: MON0800RMXH (Team 1-RSW2S3G5) Subfolder named as "Member Name" to put an individual works. Report File naming format: TeamName (TeamNo - ProgrammeSemGroup) - StudentFullNamesWithAlphabeticalOrder Example: MON0800RMXH (Team 1-RSW2S3G5) -Alice Ng-Bob Lee-Cat Tan-Dickson Do NOT upload in zip file.	Headers and Footers	Appropriate footers and headers should be used to enhance clarity and presentation.	Page Numbering	Ensure that all pages (except cover page) are numbered.	Paper Size	A4 (210 mm X 297 mm)
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	Note: Digital (softcopy) submission – to be submitted by the team leader and team members to the course’s Google Drive folder. Submit assignment documentation (.docx file format) and script file (.txt file format) here. At the same time, create subfolders (name for each member) and put all related items (queries, procedures, reports, triggers and others using .txt file format) in this subfolder.
Outline of Problem	Design a relational database system with the appropriate integrity constraints and security control, for a 88 Speedmart System. Besides, you should use PL/SQL to implement business rules with stored procedures and triggers for this 88 Speedmart System. Mandatory Module: 1. Member registration (free of charge for “Normal Member” and RM12.00 per year for “VIP Member”). 2. Allow members to order grocery items online or walk-in from any available branches. (Note: System also able to manage non-member purchases.) 3. Member can opt to pickup the grocery items or request delivery service from any delivery companies (e.g. Grab Grocery, Grocery Panda or others with delivery charge). 4. Manage the registration of the new branches. 5. Manage the stock of items for branches. 6. Enable advanced search options for items (e.g., search by name, price, category, promotion and others). 7. Manage point redemptions for discount and/or free delivery vouchers from members. Additional Modules (Each team is required to choose at least ONE module): 1. Promotion Management: manage promotions for grocery items. 2. Rating Management: manage the rating of grocery items and delivery services. 3. Refund/Return Management: manage refund of missing, broken, expired or wrong grocery items. 4. Member Feedback Management: manage feedback or complaint from members. 5. MyKasih Management: manage purchase items (selected items only) using IC from members or non-members (for pickup or walk-in only). Assignment Title: 88 Speedmart System (with Additional Module(s)) Example: 88 Speedmart System (with Promotion and Rating Management)

Assessment	- Higher marks will be awarded for work that link up all the requested parts instead of presenting each part as individual separate component. - Every student must be involved in every part of the assignment. Each student must produce SQL statements, procedure(s), trigger(s), report(s), do presentation and etc. Marks will be awarded individually for task 4 through 9. - Students from the same team are not to produce similar work. Thus, team discussion is necessary to prevent overlapping work from being produced. <table><tr><th>No.</th><th>Task</th><th>% of allocation</th><th>CLO</th></tr><tr><td>1</td><td>ERD</td><td>10%</td><td>1</td></tr><tr><td>2</td><td>Data Definition (DDL)</td><td>10%</td><td>1</td></tr><tr><td>3</td><td>Data records</td><td>5%</td><td>1</td></tr><tr><td>4</td><td>Queries</td><td>10%</td><td>3</td></tr><tr><td>5</td><td>Stored Procedures</td><td>10%</td><td>2</td></tr><tr><td>6</td><td>Triggers</td><td>10%</td><td>2</td></tr><tr><td>7</td><td>Reports</td><td>20%</td><td>3</td></tr><tr><td>8</td><td>Extra Efforts</td><td>10%</td><td>2</td></tr><tr><td>9</td><td>Presentation and Participation</td><td>15%</td><td>3</td></tr><tr><td></td><td>Total</td><td>100%</td><td></td></tr></table> Refer to the Assignment Assessment Form for more detail on the assessment criteria.	No.	Task	% of allocation	CLO	1	ERD	10%	1	2	Data Definition (DDL)	10%	1	3	Data records	5%	1	4	Queries	10%	3	5	Stored Procedures	10%	2	6	Triggers	10%	2	7	Reports	20%	3	8	Extra Efforts	10%	2	9	Presentation and Participation	15%	3		Total	100%	
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What you should hand in	The following items are to be handed in by each team: - A cover page and the Coursework Declaration. Refer to Sample Report Content.docx. - A softcopy of the final group report which must include all the work done for Task 1 through 8, and the outputs of the execution of queries, stored procedures, triggers and reports. Explanation on the need and the importance of the queries, stored procedures, triggers and reports created for the company should also be included. Refer to the Assignment Submission Template for more detail on the final report. <u>IMPORTANT</u> Submitting the assignment means you have agreed that your work is original and comply with the rules and regulations (refer to Academic Impropriety)																																												
Estimated Time	At least 18 hours per team member.																																												
Academic Impropriety	You may only work with the students in your team to produce your deliverables for this assignment. Works submitted must be ORIGINAL. You can discuss with your friends. You can research from web resources. Do not share your work with other teams.																																												

	Your attention is drawn to the Faculty of Computing and Information Technology Handbook and the University College guidelines on avoiding plagiarism. This covers cheating, attempts to cheat, plagiarism, collusion and any other attempts to gain an unfair advantage in assessment. The work that you submit must conform to those regulations.
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No	Task	Deadline
1.	A complete ER data model in 3rd Normal Form. All assumptions and business rules must be clearly written and presented. All primary keys, foreign keys, relationships (weak and strong relationships) and attributes must be clearly shown. Practical lecturer will check on students' ERD (exclude attributes) start from week 2 onward.	Week 2-4
2.	Prepare the Data Definition Language (DDL) statements to implement tables for the system using the Oracle database system. The following must be included in the DDL statements: Choose appropriate data types (2 marks), default values (1 mark), check constraints (2 marks) and NOT NULL clauses (1 mark). Ensure that you enforce the Entity Integrity Rule (2 marks) and the Referential Integrity Rule (2 marks).	Week 5
3.	You are advised to insert sufficient data records for each table that you have created. Write the appropriate INSERT statements to populate your database. Enter at least 10 records for each base (parent) table, at least 100 records for each transaction (child) table and at least 300 records for each associative (bridge) table. Transaction tables would have more records than base tables. For example, if there are 10 customers, average of 2 order per customer per year with average of 2 items per order. Customer table: 10 records (base table) Orders table: $10 * 2 = 20$ records per year (transaction table) OrderDetails table: $20 * 2 = 40$ records per year (transaction) Furthermore, you should create sample data that has different date/month/year to simulate a real-world environment. Website to generate random records: https://www.mockaroo.com/ * Run the single script file from Task 2 and 3 to create the system database on week 8 to 9.	Week 6-7

4.	Produce queries to extract relevant information for decision making. Single table queries are not allowed. Make use of multiple tables and aggregate functions where necessary. Judge on usefulness and complicity of each queries. Views to be incorporated where appropriate. Each student is to produce 2 queries. The queries produced by the team must address the informational needs of management at any levels: Strategic, Tactical and Operational. Single table queries are not allowed.	Week 8-10
5.	Design and create stored procedures that cater for the various use case scenarios for the system. Each student is to produce 2 stored procedures. Procedures can be used for data retrieval (Selection with/without cursor), data manipulation (Add, Update, Delete), validation, etc.	
6.	Design and create triggers that enforces system-wide business rules and policies. Each student is to produce 2 triggers.	
7.	Write procedures to generate reports (include summary, detail and on demand basis reports) for the organization. On demand basis report is a report that will be generated once it is called by the users, and parameter value(s) might be passed in during the call. Each student is to produce 2 reports (either detail, summary or demand report). Use cursor in report generation, nested cursor is more preferable. a) Single cursor: 6 marks +- b) Multiple cursor: 7 marks +- c) Nested cursor: 8 marks +-	
8.	Each student incorporate 1 sequence, 2 views (for both queries), 2 indexes, 2 output formatting (for both queries), 2 exceptions (2 different types: RAISE_APPLICATION_ERROR, default exception, custom exception or PRAGMA EXCEPTION_INIT) and/or functions when necessary. (2 marks for each correct extra effort item.)	
9.	Presentation and Participation • Demonstration and explanation on all the tasks (Task 4 - 8) • Q & A • Active participation in class discussion.	Week 11-14